Sri Lanka Institute of Information Technology



Data Warehousing and Business Intelligence

IT3021

Year 3, Semester 1, 2026

Assignment 02

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1.Data Warehouse Design

The Accident Data Warehouse (DW) developed in Assignment 01 is used as the data source for this assignment. The warehouse follows a **star schema** design, where the central fact table FactAccident stores measurable data such as the number of persons injured and killed.

The fact table is connected to dimension tables that provide descriptive context. The DimDate table enables time-based analysis using attributes such as Year, Month, and Day. The DimLocation table contains geographical information such as Borough, Zip Code, and street names.

The DimLocation dimension is implemented as a **Slowly Changing Dimension Type 2 (SCD Type 2)**, allowing the system to track historical changes in location attributes.

This structure supports multidimensional analysis and enables effective reporting using SSAS, Excel, and Power BI.

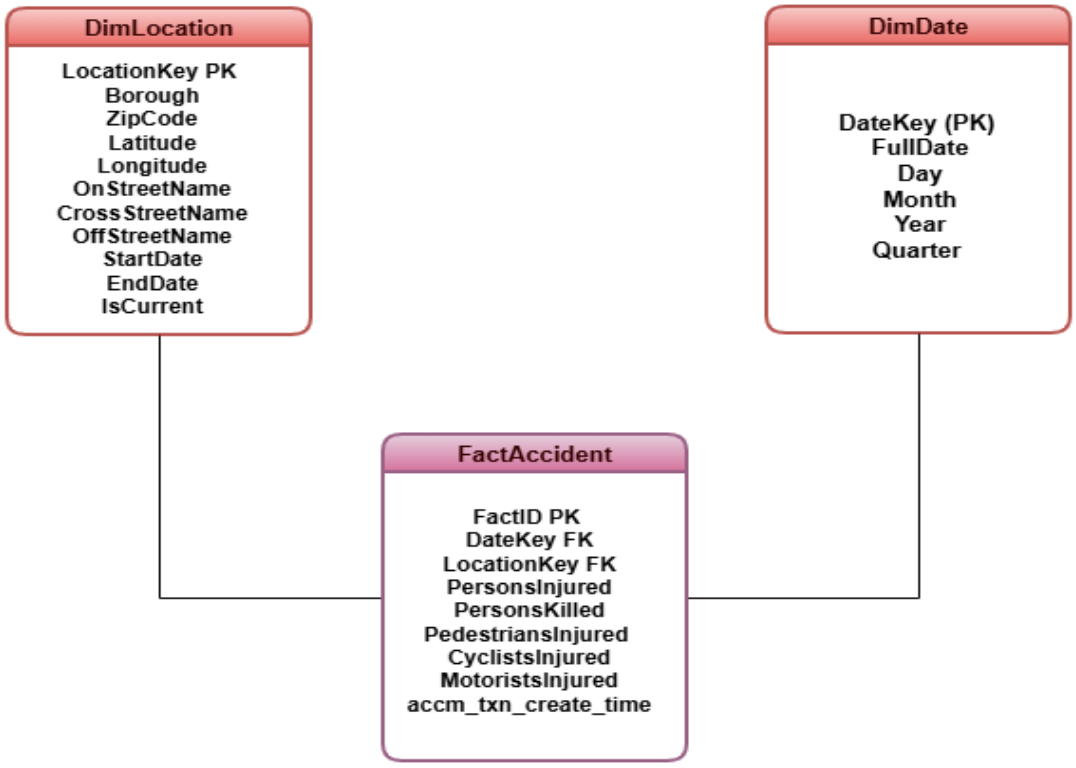


Figure 1: Star Schema of Accident Data Warehouse

2.SSAS Cube implementation

1. **Created SSAS Project**
   A new Analysis Services Multidimensional Project named Accident_Cube was created using Visual Studio.

2. **Established Data Source**
   A connection to the Accident_DW database was configured using Windows authentication.

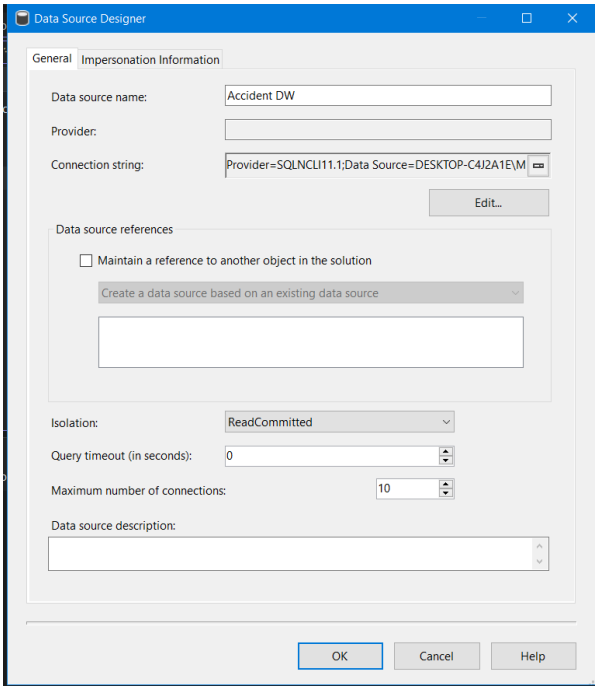


Figure 2: Data Source Configuration

3. **Built Data Source View (DSV)**
   The tables FactAccident, DimDate, and DimLocation were added and relationships were verified.

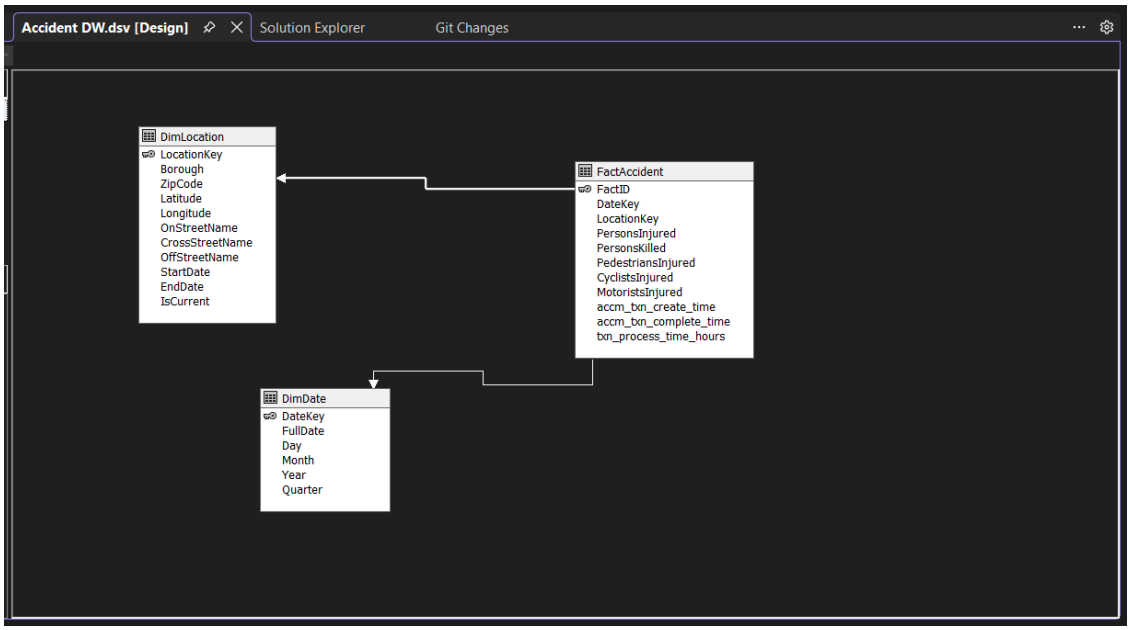


Figure 3: Data Source View with Relationships

4. **Created Cube**
   The Cube Wizard was used with the "Use existing tables" option, selecting FactAccident as the measure group.

5. **Added Dimensions**
   The dimensions DimDate and DimLocation were added to the cube.

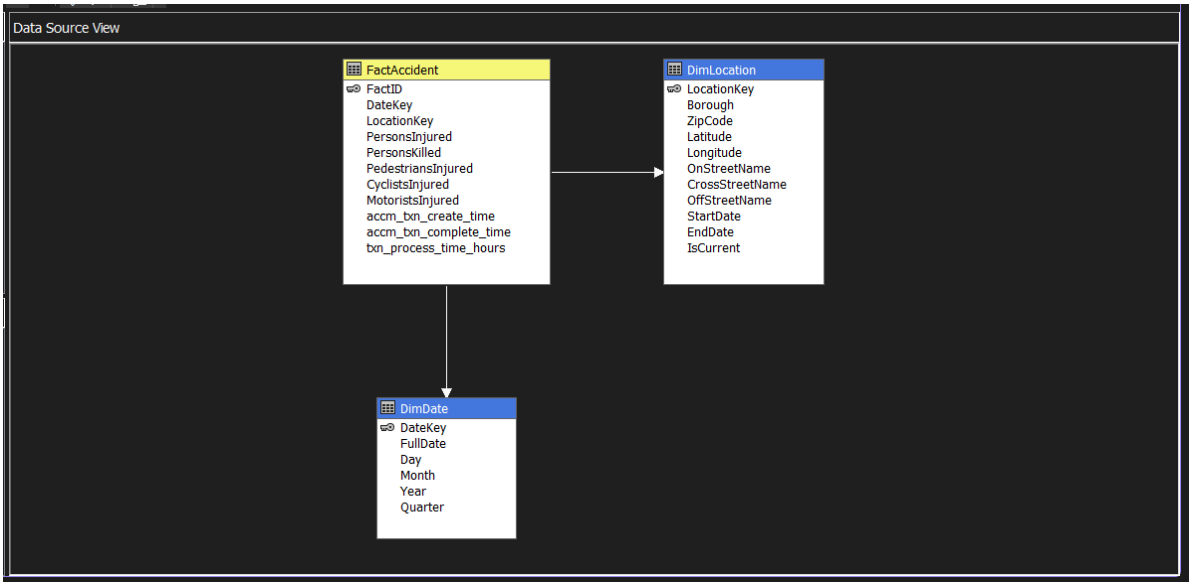


Figure 4: Cube Structure

6. **Created Hierarchies**

- Date Hierarchy → Year → Month → Day
- Location Hierarchy → Borough → OnStreetName

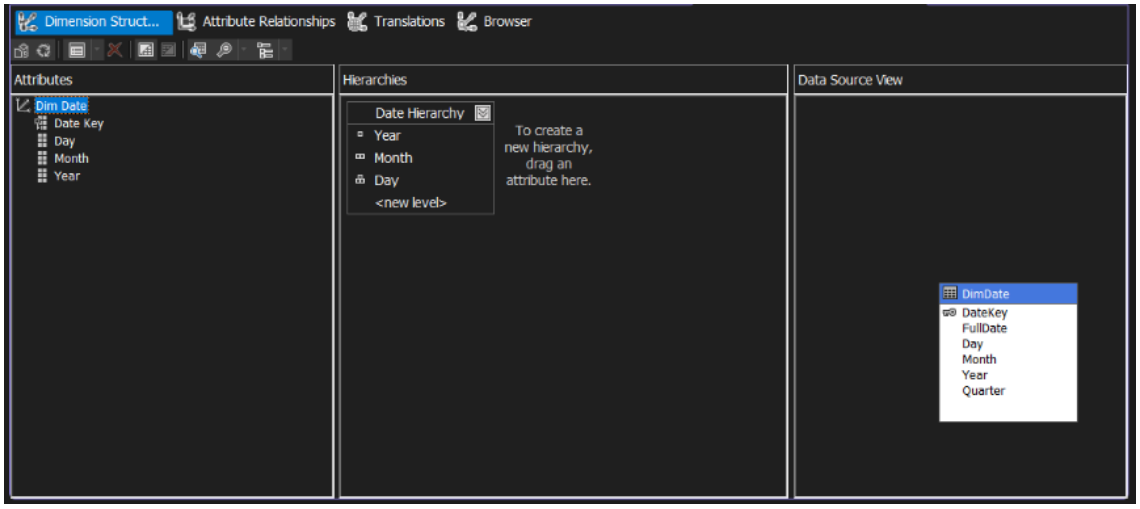


Figure 5: Date Hierarchy

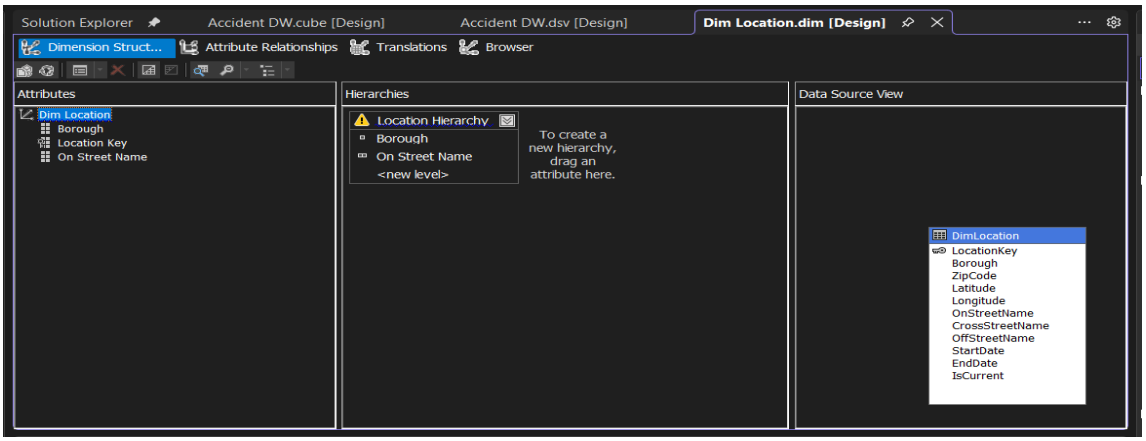


Figure 6: Location Hierarchy

7. **Deployed and Processed Cube**
   The cube was successfully deployed and processed in the local SSAS server.

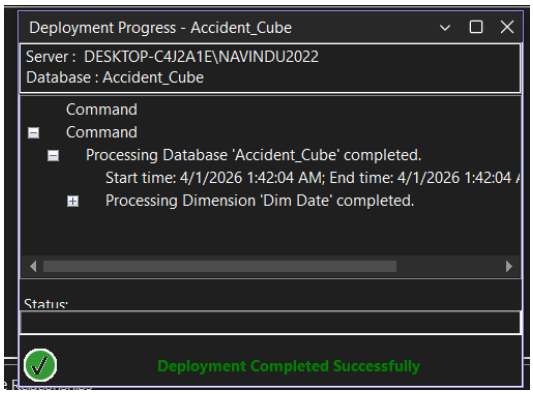


Figure 7: Cube Deployment Success

3.Demonstration of OLAP operations

1. **Connected Excel to SSAS Cube**
   Excel was connected to the deployed cube using the Analysis Services connection.

2. **Created PivotTable**
   A PivotTable was created to perform multidimensional analysis.

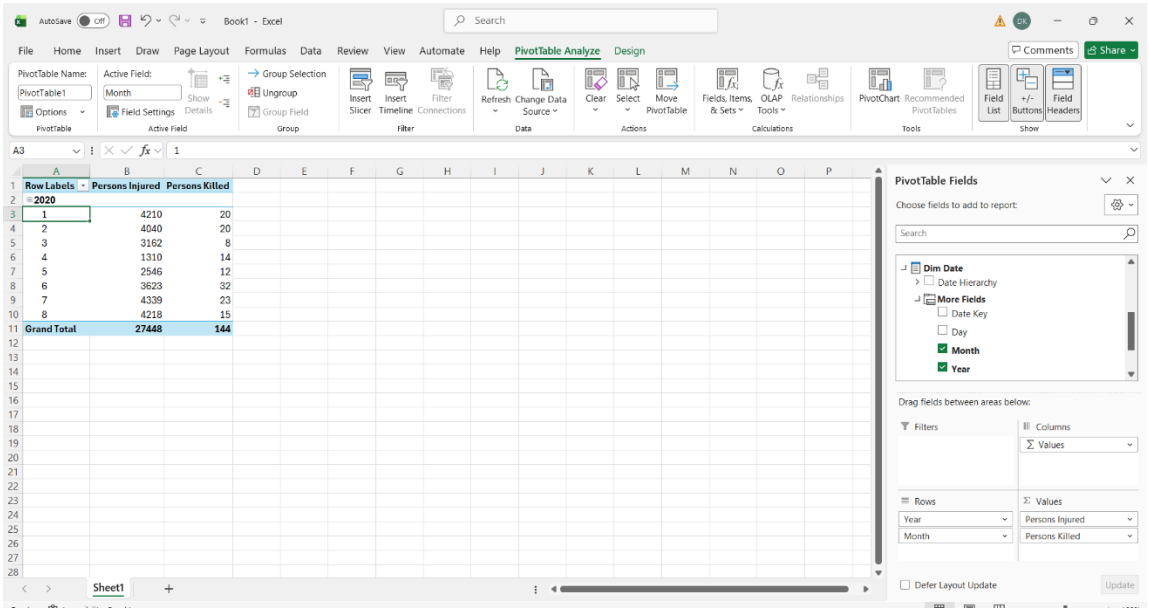
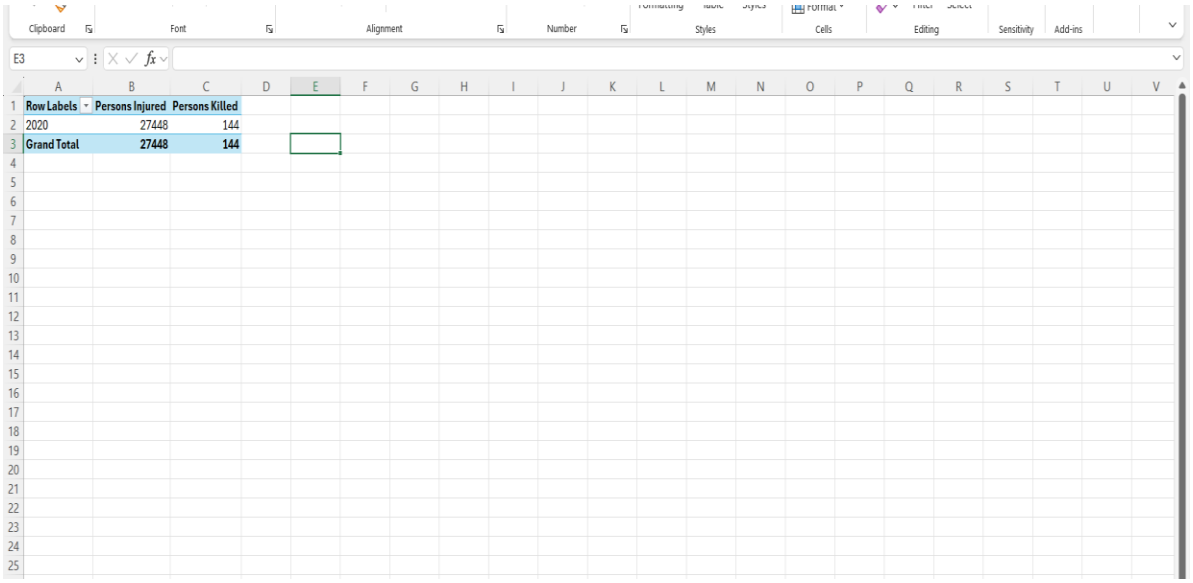


Figure 8: PivotTable connected to SSAS cube

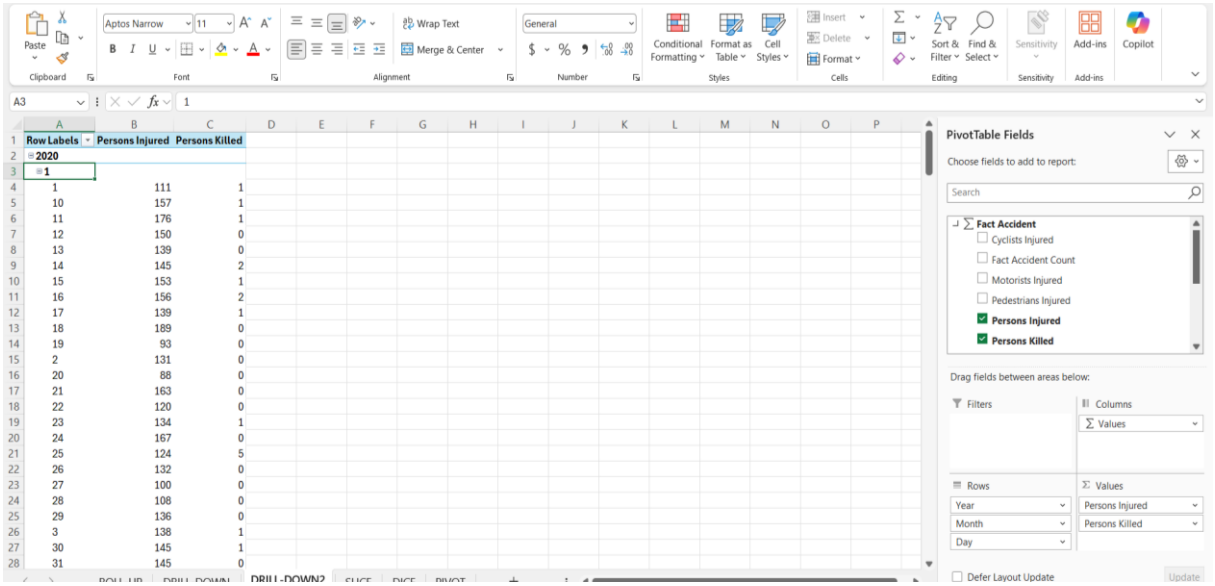
3. **Roll-up Operation**
   Data was aggregated from lower levels to higher levels (e.g., monthly data to yearly totals).



Row Labels	Persons Injured	Persons Killed
2020	27448	144
Grand Total	27448	144

Figure 9: Roll-up and Drill-down operations

4. **Drill-down Operation**
   The Date Hierarchy was used to explore detailed levels from Year to Month to Day.

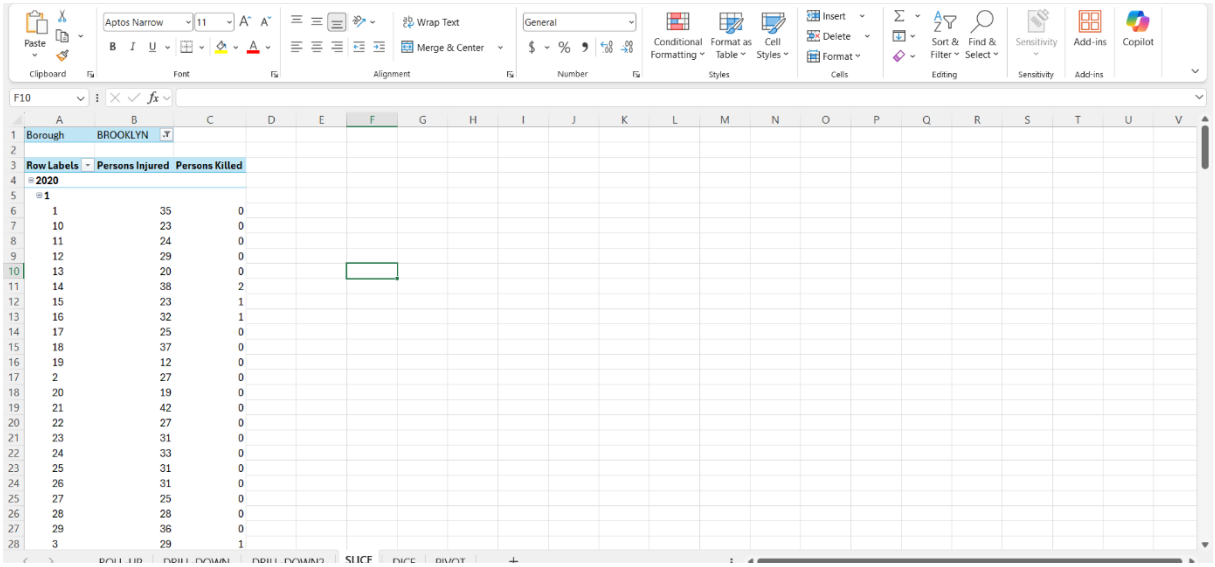


Row Labels	Persons Injured	Persons Killed
2020		
1		
1	111	1
10	157	1
11	176	1
12	150	0
13	139	0
14	145	2
15	153	1
16	156	2
17	139	1
18	189	0
19	93	0
2	131	0
20	88	0
21	163	0
22	120	0
23	134	1
24	167	0
25	124	5
26	132	0
27	100	0
28	108	0
29	136	0
3	138	1
30	145	1
31	145	0

Figure 10: Roll-up and Drill-down operations

5. **Slice Operation**

   Data was filtered using a single condition (e.g., Borough = Bronx).

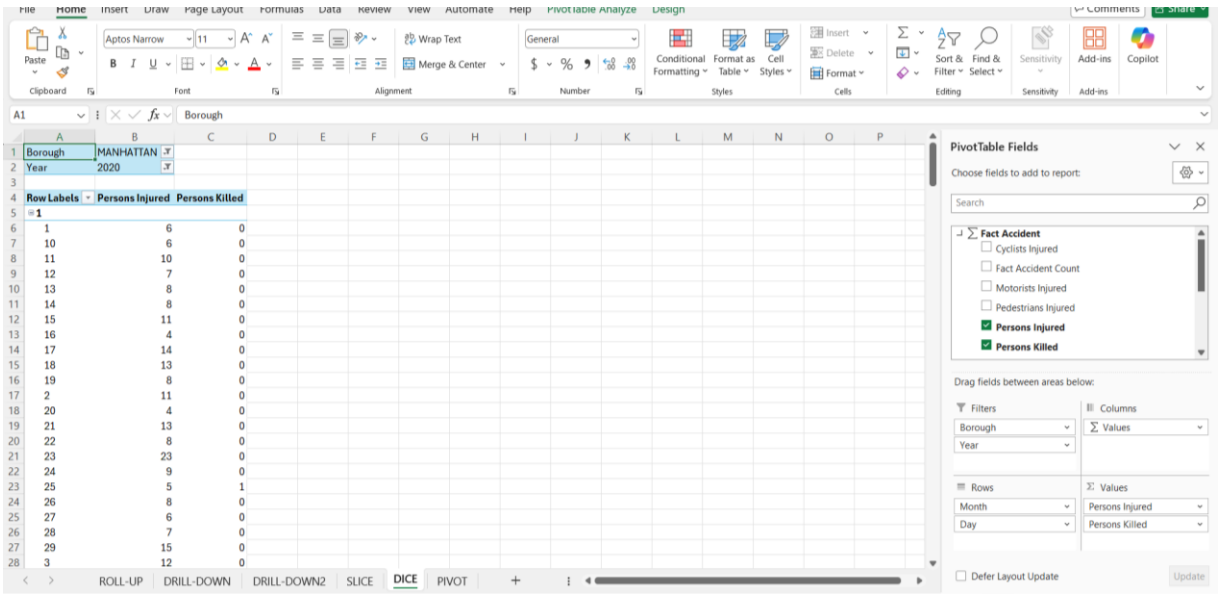


Borough	BROOKLYN	
Row Labels	Persons Injured	Persons Killed
2020		
1		
1	35	0
10	23	0
11	24	0
12	29	0
13	20	0
14	38	2
15	23	1
16	32	1
17	25	0
18	37	0
19	12	0
2	27	0
20	19	0
21	42	0
22	27	0
23	31	0
24	33	0
25	31	0
26	31	0
27	25	0
28	28	0
29	36	0
3	29	1

Figure 11: Slice operation using Borough filter

6. **Dice Operation**

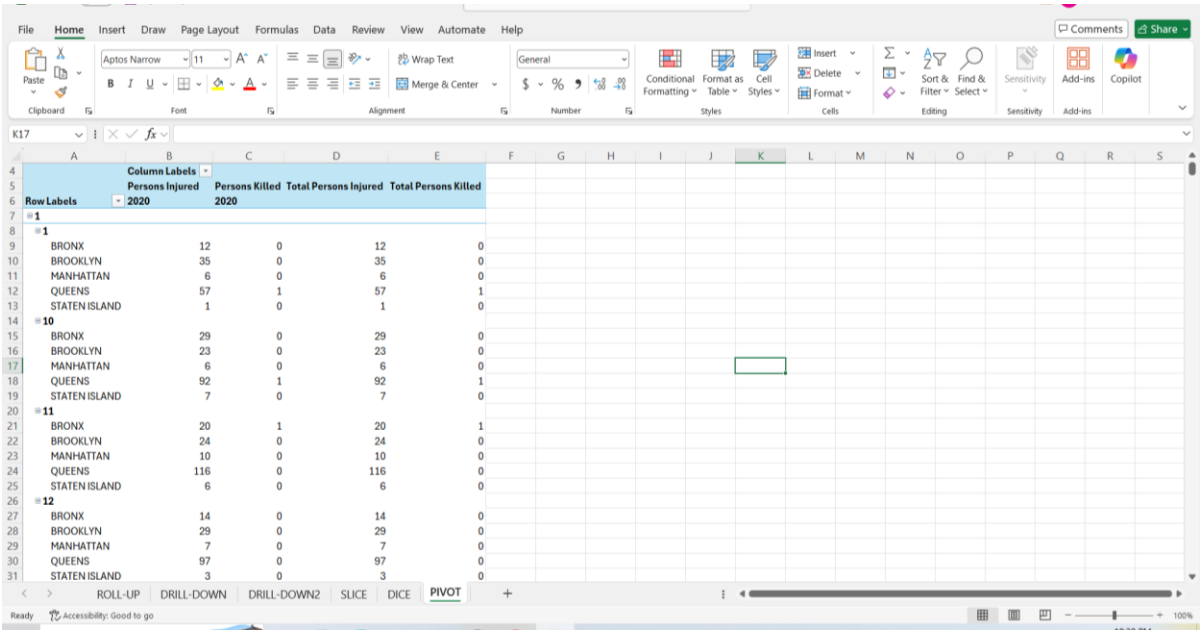
   Multiple filters were applied simultaneously (e.g., Borough = Bronx and Year = 2020).



Borough	MANHATTAN	
Year	2020	
Row Labels	Persons Injured	Persons Killed
1		
1	6	0
10	6	0
11	10	0
12	7	0
13	8	0
14	8	0
15	11	0
16	4	0
17	14	0
18	13	0
19	8	0
2	11	0
20	4	0
21	13	0
22	8	0
23	23	0
24	9	0
25	5	1
26	8	0
27	6	0
28	7	0
29	15	0
3	12	0

Figure 12: Dice operation using multiple filters

7. **Pivot Operation**
   Rows and columns were rearranged to view data from different perspectives.



Row Labels	Column Labels: Persons Injured 2020	Persons Killed 2020	Total Persons Injured	Total Persons Killed
1				
1				
BRONX	12	0	12	0
BROOKLYN	35	0	35	0
MANHATTAN	6	0	6	0
QUEENS	57	1	57	1
STATEN ISLAND	1	0	1	0
10				
BRONX	29	0	29	0
BROOKLYN	23	0	23	0
MANHATTAN	6	0	6	0
QUEENS	92	1	92	1
STATEN ISLAND	7	0	7	0
11				
BRONX	20	1	20	1
BROOKLYN	24	0	24	0
MANHATTAN	10	0	10	0
QUEENS	116	0	116	0
STATEN ISLAND	6	0	6	0
12				
BRONX	14	0	14	0
BROOKLYN	29	0	29	0
MANHATTAN	7	0	7	0
QUEENS	97	0	97	0
STATEN ISLAND	3	0	3	0

Figure 13: Pivot operation

4.Power BI Reports

Report 1: Matrix Report

A Matrix visual was created to display tabular data with grouped rows and columns. The Borough attribute was used as rows, and Year was used as columns, while Total Injured and Total Killed were used as values.

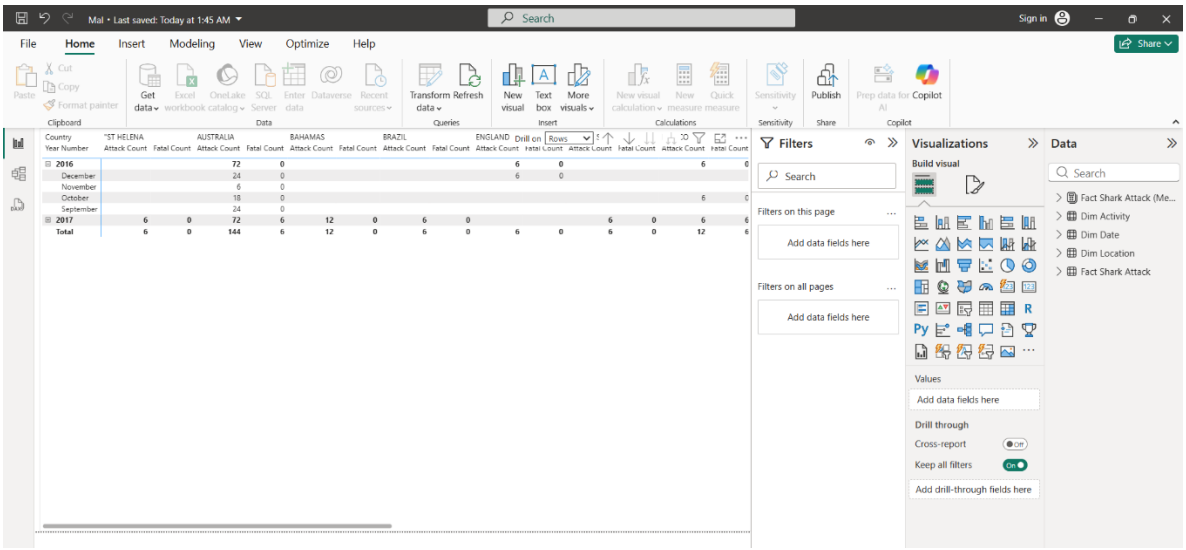


Figure 14: Matrix report showing accidents by Borough and Year

Report 2: Slicers and Interactive Report

A report was created with multiple slicers to enable interactive filtering. Two slicers were used:

- Borough slicer
- Year slicer

Selecting a value in one slicer dynamically filters the data in other visuals, demonstrating cascading filter behavior.

Multiple visuals were included in this report, such as:

- Bar chart (injuries by borough)
- Line chart (trend over time)

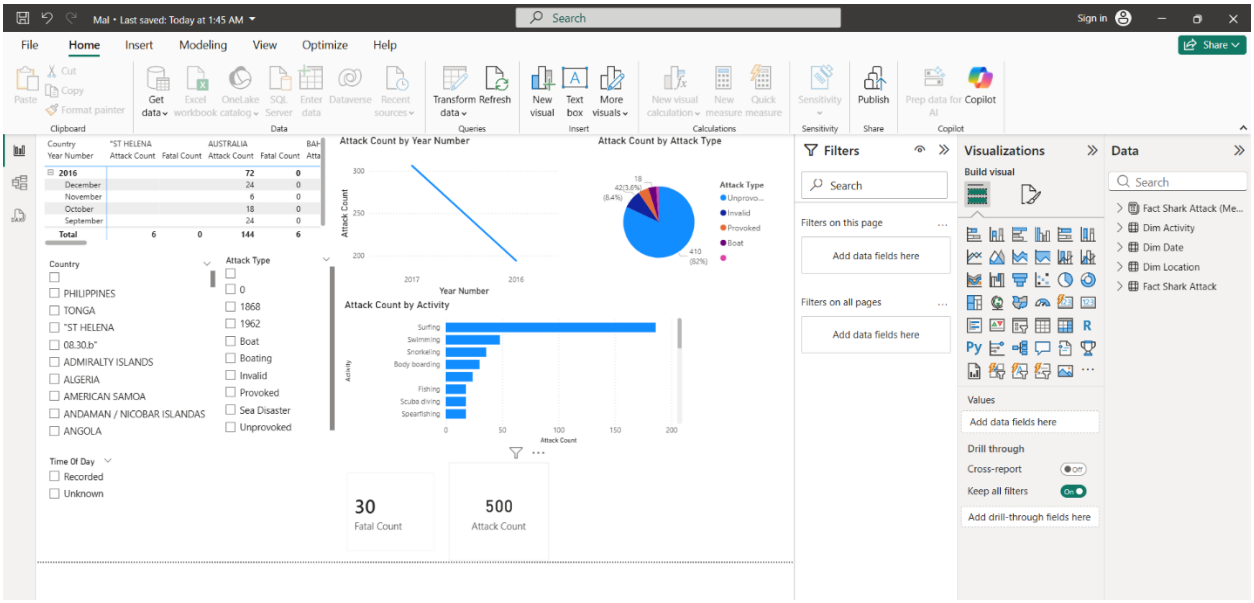


Figure 15: Report with cascading slicers and interactive visuals

Report 3: Drill-down Report

A drill-down report was created using a line chart with a Date Hierarchy. Users can explore data at different levels:

- Year → Month → Day

The drill-down functionality allows users to analyze trends at detailed levels.

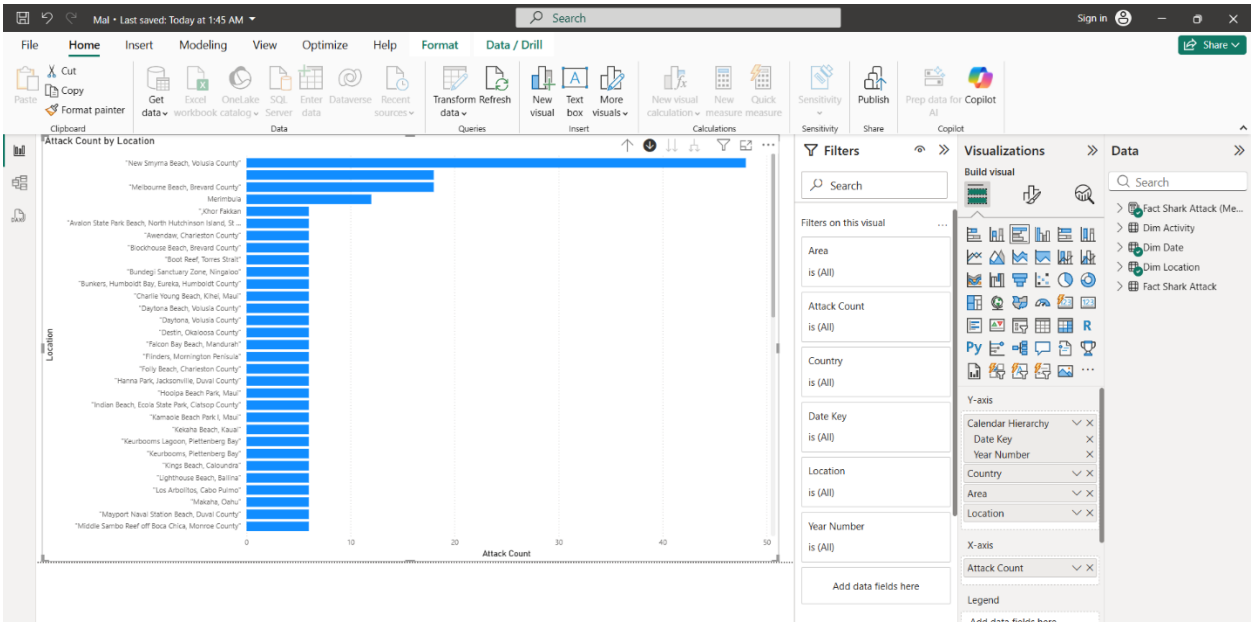


Figure 16: Drill-down analysis using Date Hierarchy

Report 4: Drill-through Report

A drill-through report was created to allow users to navigate to a detailed page.

A separate page was created, and the Borough field was added as a drill-through filter. Users can right-click on a visual and navigate to this page to view detailed data related to the selected Borough.

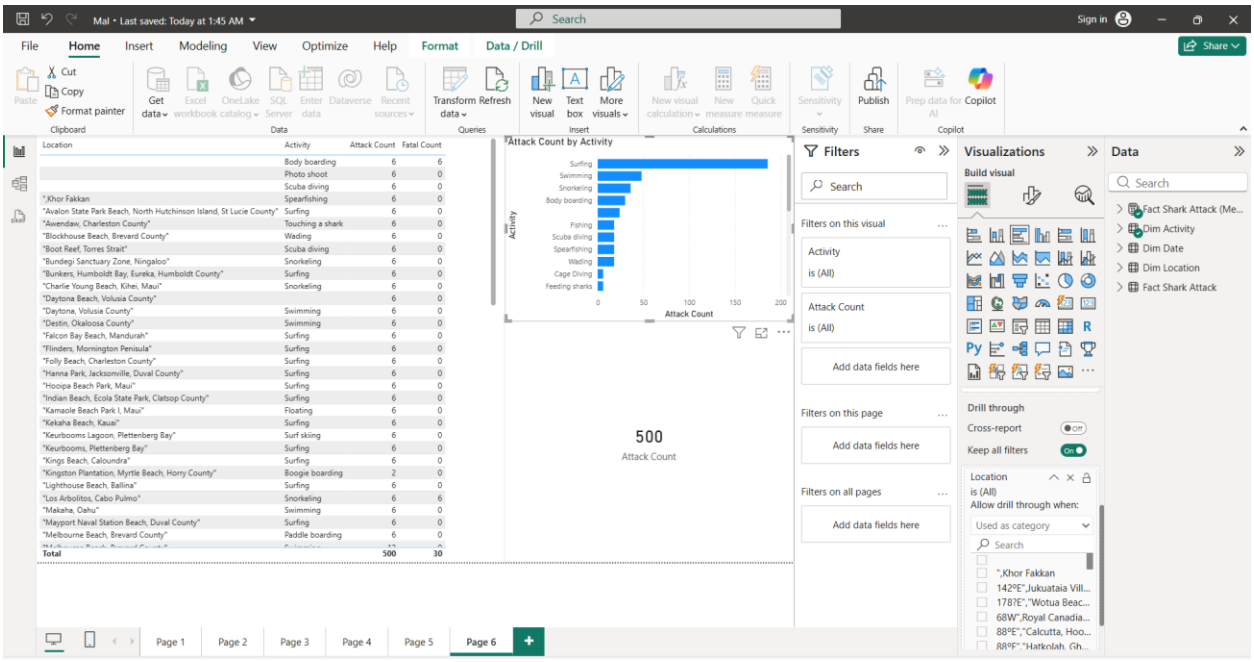


Figure 17: Drill-through report showing detailed information

5.DAX Measures used for Accident Analysis

Total Injured

Total Injured = SUM(FactAccident[PersonsInjured])

Total Killed

Total Killed = SUM(FactAccident[PersonsKilled])

6.Conclusion

This assignment successfully demonstrated the implementation of a multidimensional cube using SSAS and the use of OLAP operations in Excel. The cube enabled efficient analysis through roll-up, drill-down, slice, dice, and pivot operations.

Furthermore, Power BI was used to create interactive reports including matrix, slicers, drill-down, and drill-through reports. These visualizations provide meaningful insights into accident trends, locations, and severity.

Overall, the assignment highlights the importance of Business Intelligence tools in transforming data into actionable insights for decision-making.